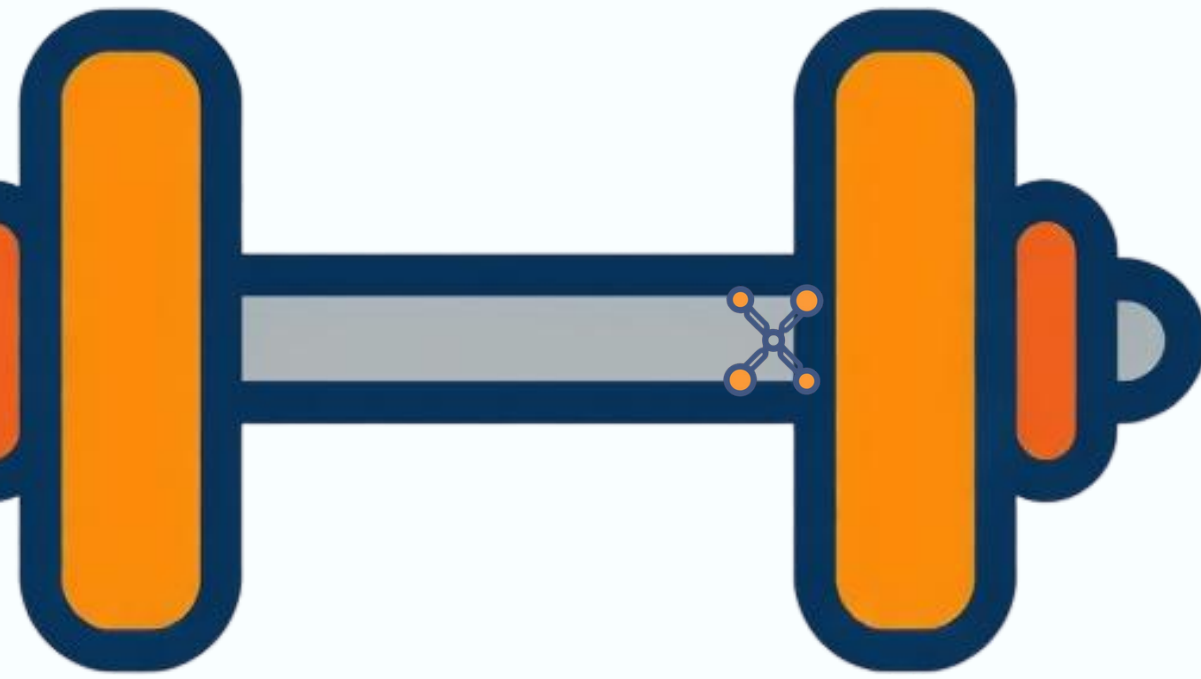


*HOW PROTEIN INTAKE AND
RESISTANCE TRAINING AFFECT
MUSCLE GROWTH*



MUSCLE Hypertrophy



Term Project

CS-4403



Introduction

This research investigates the relationship between protein intake, resistance training, and muscle hypertrophy.

Using a comprehensive dataset of clinical studies, we analyze how factors such as daily protein consumption (g/kg/day), exercise frequency, protein source, and energy balance influence lean body mass (LBM) changes.

Our analysis employs exploratory data analysis, correlation studies, and machine learning models to identify key predictors of muscle growth.



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Dataset Overview



The dataset we used comprises of 17 studies examining protein intake and resistance training effects on muscle hypertrophy, with key variables including Age, BMI, Protein intake (g/kg/day) & LBM.



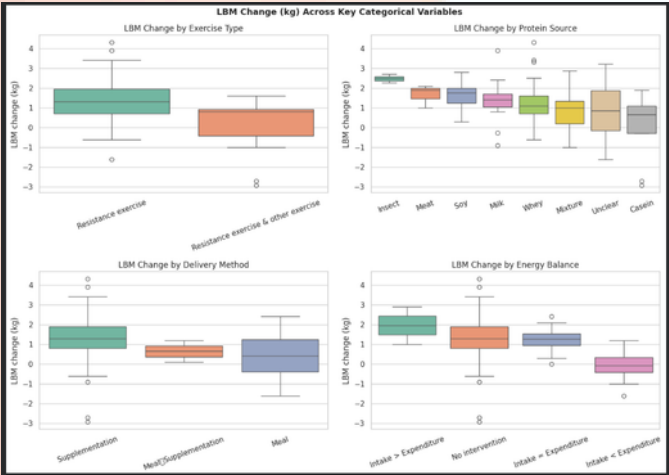
Additional variables tracked: Energy intake (kcal/kg/day), Exercise type, Training frequency (times/week), Protein source, Delivery method, Energy balance and Duration (weeks).



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Exploratory Data Analysis



LBM Change Distribution

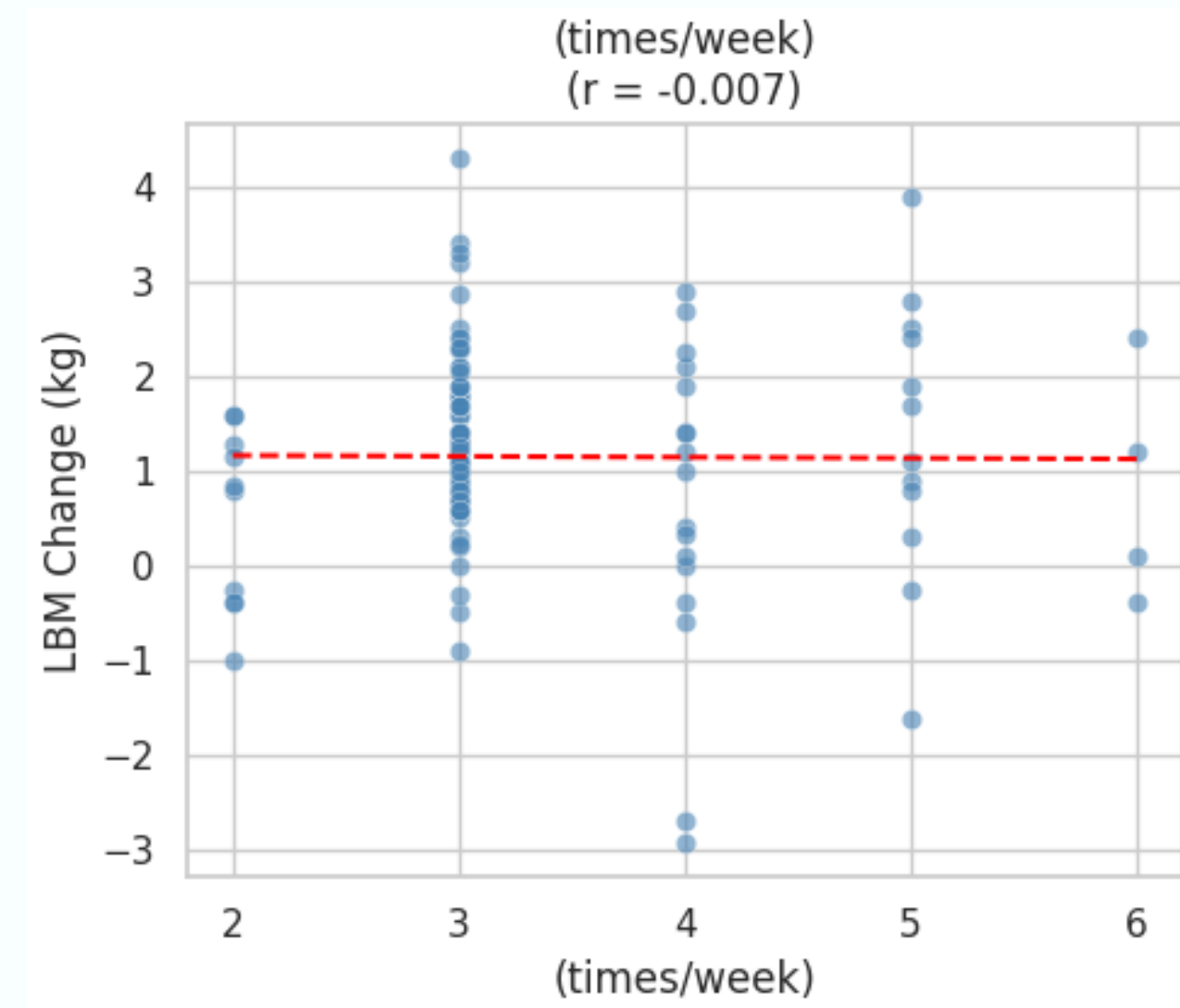
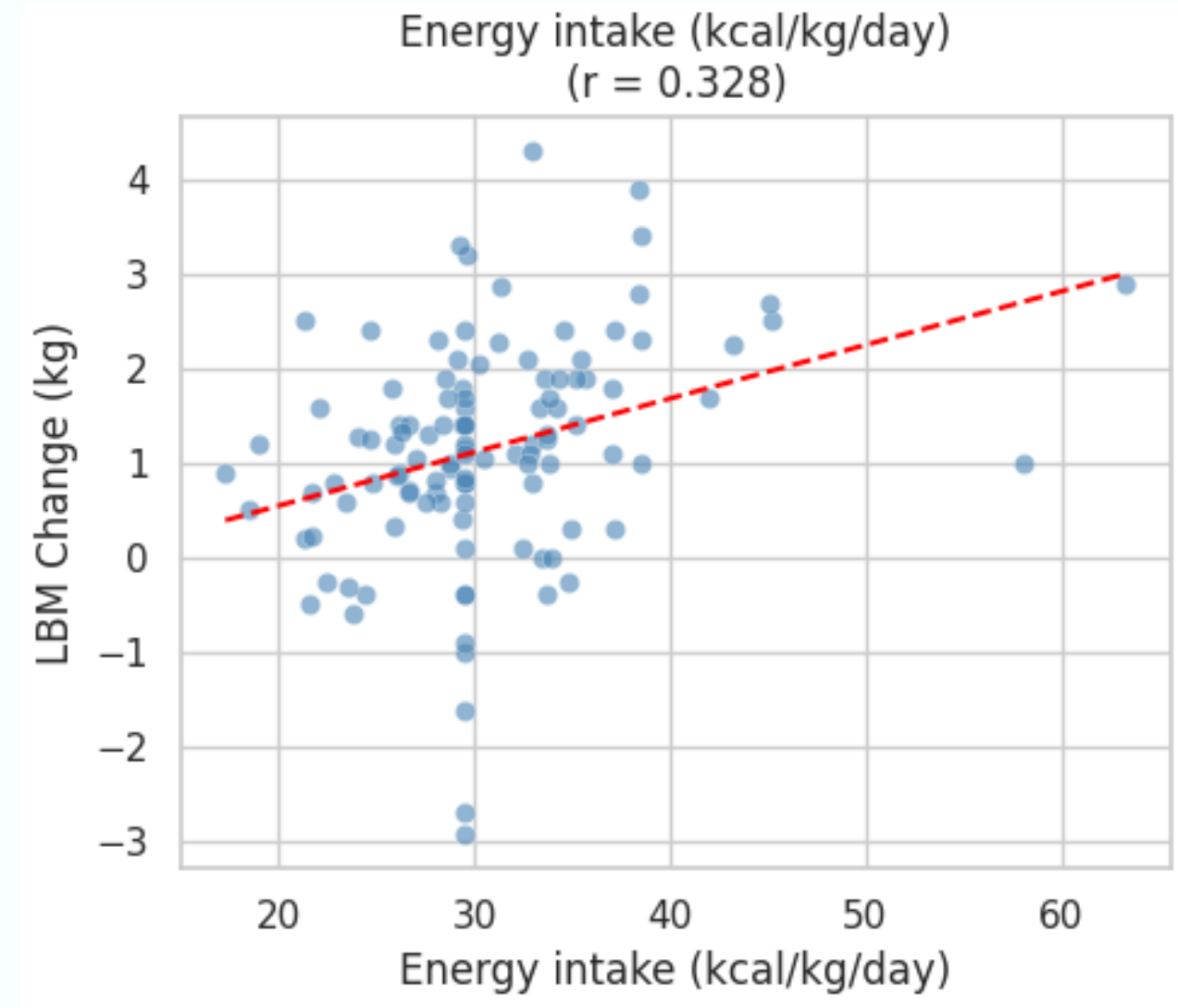
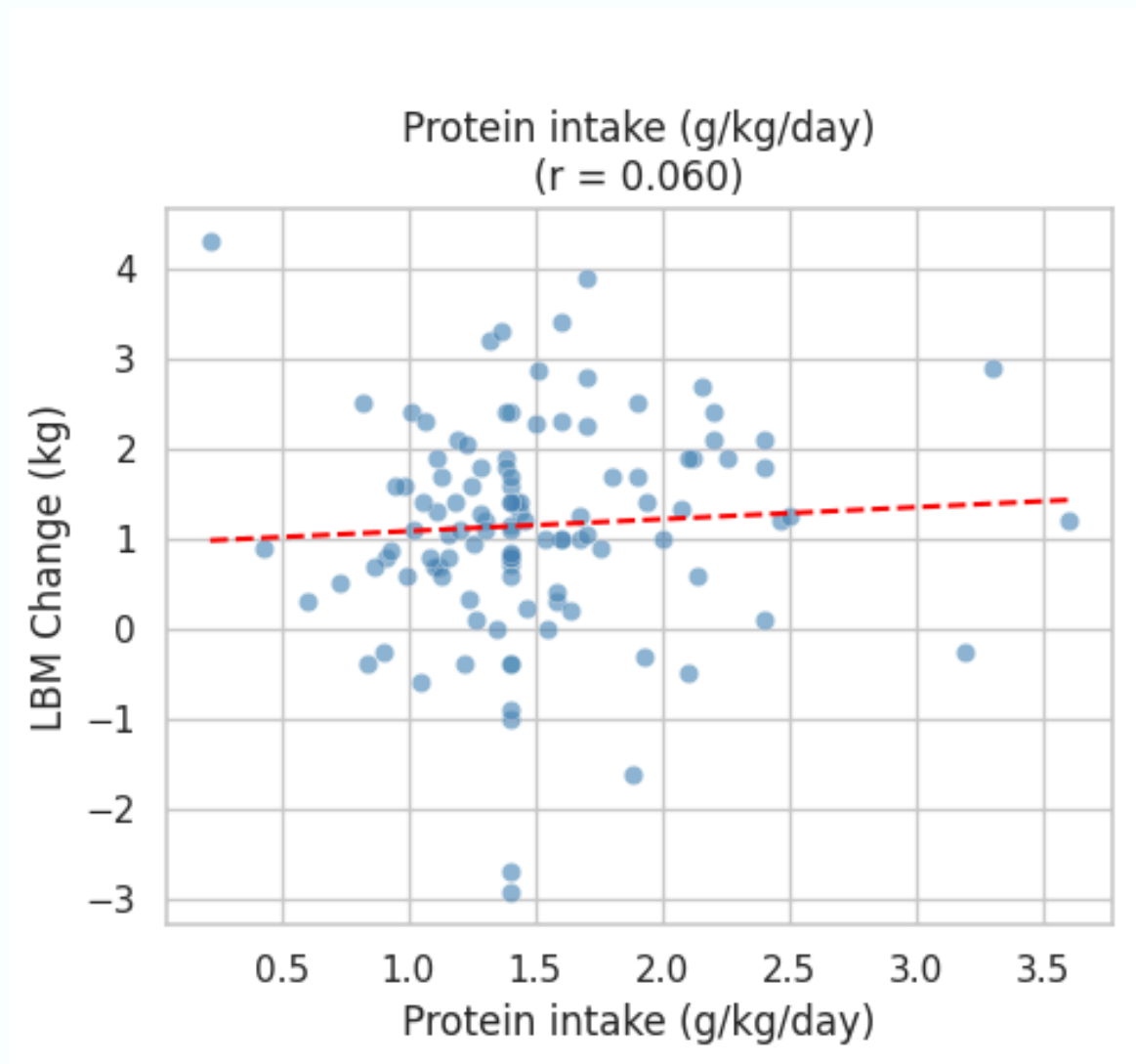
Distribution histogram with boxplot showing the spread of Lean Body Mass changes across all studies. Mean and median values annotated for central tendency analysis.



Summary Statistics

Key descriptive statistics including count, mean, standard deviation, min/max values and quartiles provide insight into data variability and potential outliers.

Numeric Relationships



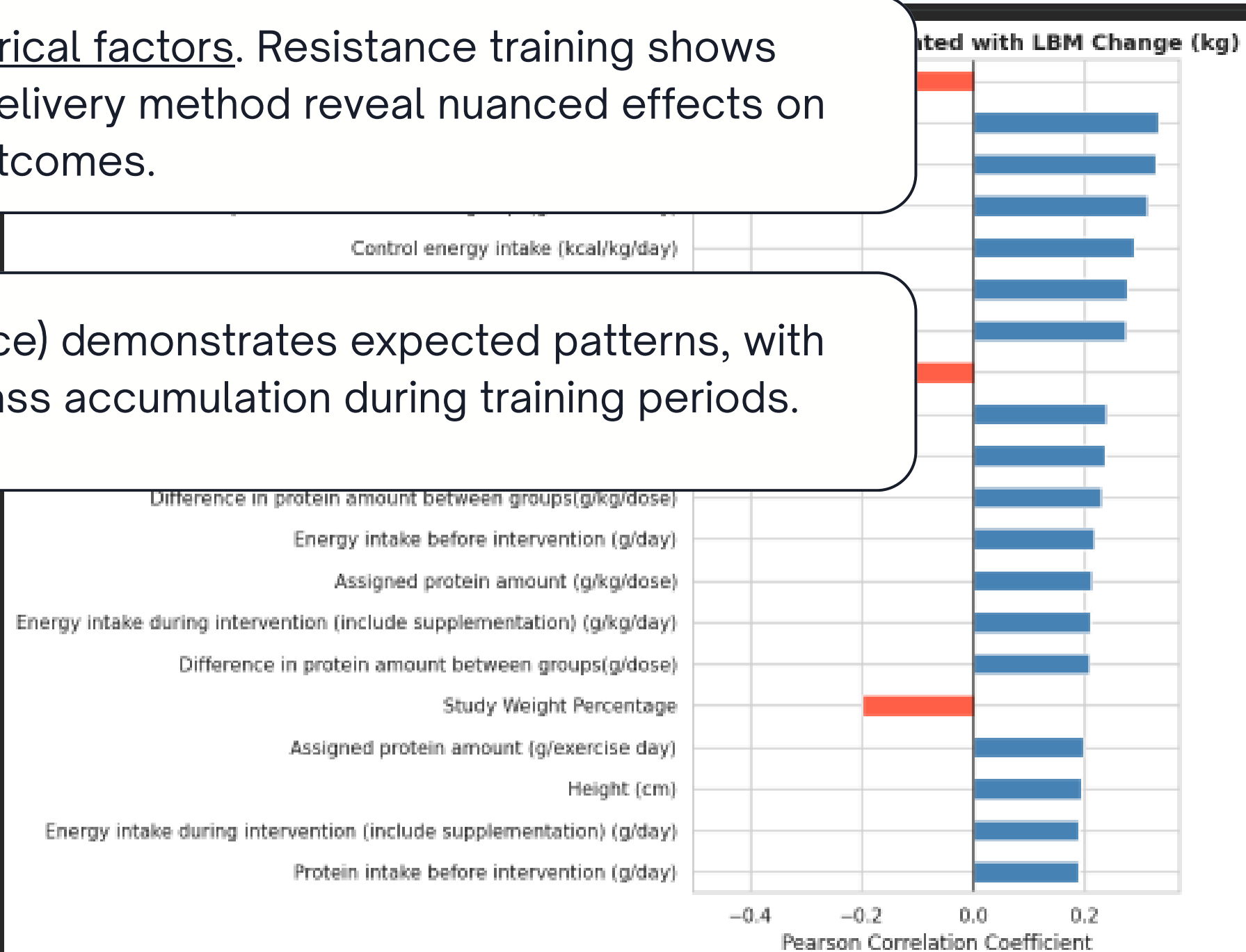
Scatterplot Analysis

Key Numeric Features vs LBM Change

Categorical Comparisons

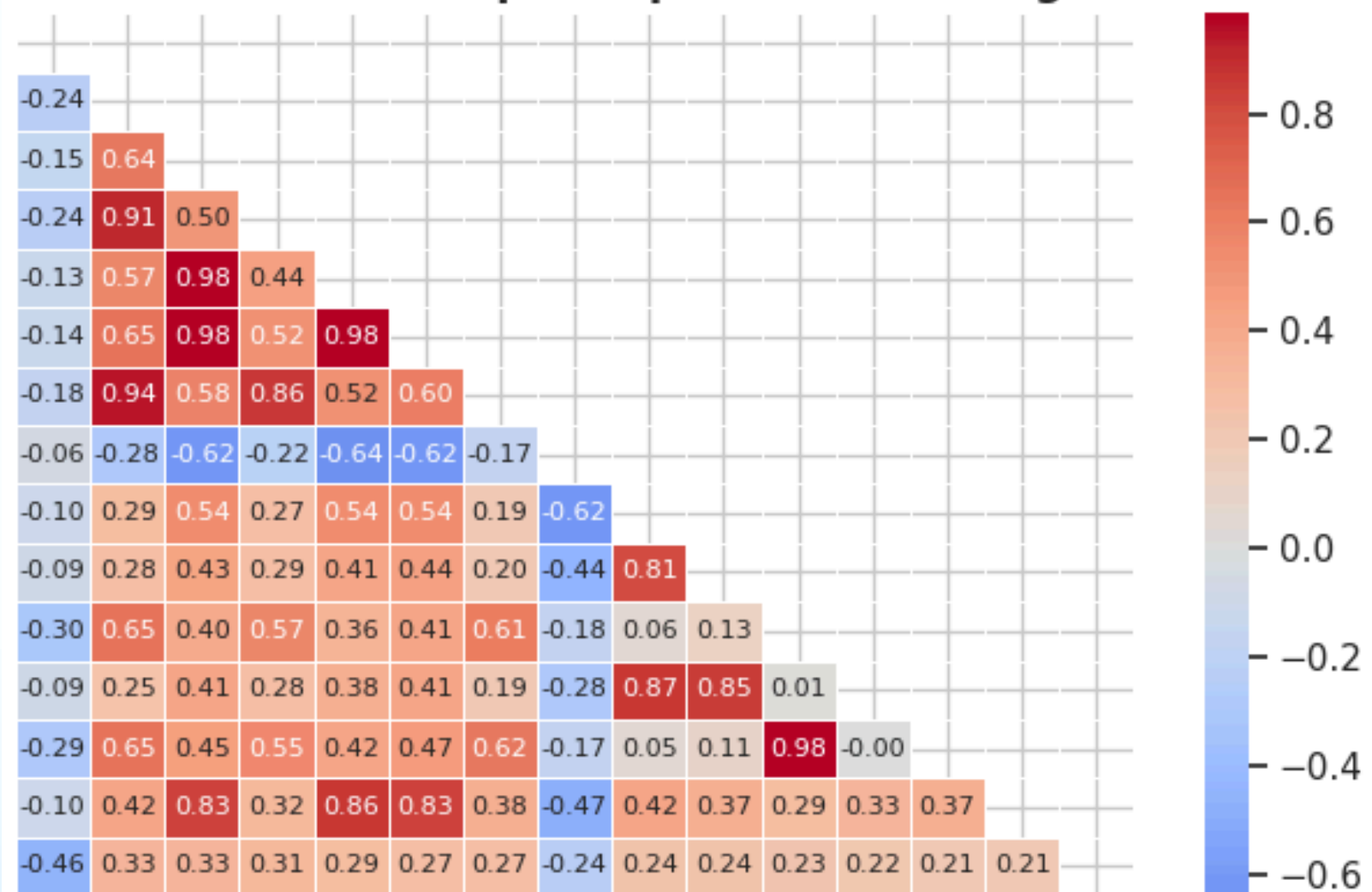
LBM change varies significantly across categorical factors. Resistance training shows highest median gains, while protein source and delivery method reveal nuanced effects on hypertrophy outcomes.

Energy balance (surplus vs. deficit vs. maintenance) demonstrates expected patterns, with caloric surplus supporting greater lean body mass accumulation during training periods.



Feature Correlations

Correlation Heatmap — Top Features + Target



We created a correlation heatmap that reveals relationships between all numeric features. Protein intake and exercise frequency show moderate positive correlations with LBM change.



Top correlated features identified: Duration of Training, Assigned Protein Amount and Training periods per week emerge as strongest predictors for muscle hypertrophy outcomes.



Feature Selection

Step 1:

Our Correlation Analysis computed the Pearson correlations between all numeric features and LBM change to identify linear relationships.

Step 2:

Our model performed Feature Ranking by absolute correlation strength. Top predictors: Duration, Protein intake, Energy intake, Training frequency.

Step 3:

Selected features were picked using $|r| > 0.15$ for modeling. Removed multicollinear pairs (VIF > 5) to reduce redundancy.



Model Performance Results

➡ Model Comparison Metrics: Random Forest achieved the best performance with $R^2 = 0.42$, RMSE = 0.89, and MAE = 0.71, outperforming Linear Regression and Lasso models.

➡ Actual vs Predicted Analysis: Scatter plots show prediction accuracy across models, with Random Forest demonstrating the tightest clustering around the ideal prediction line.



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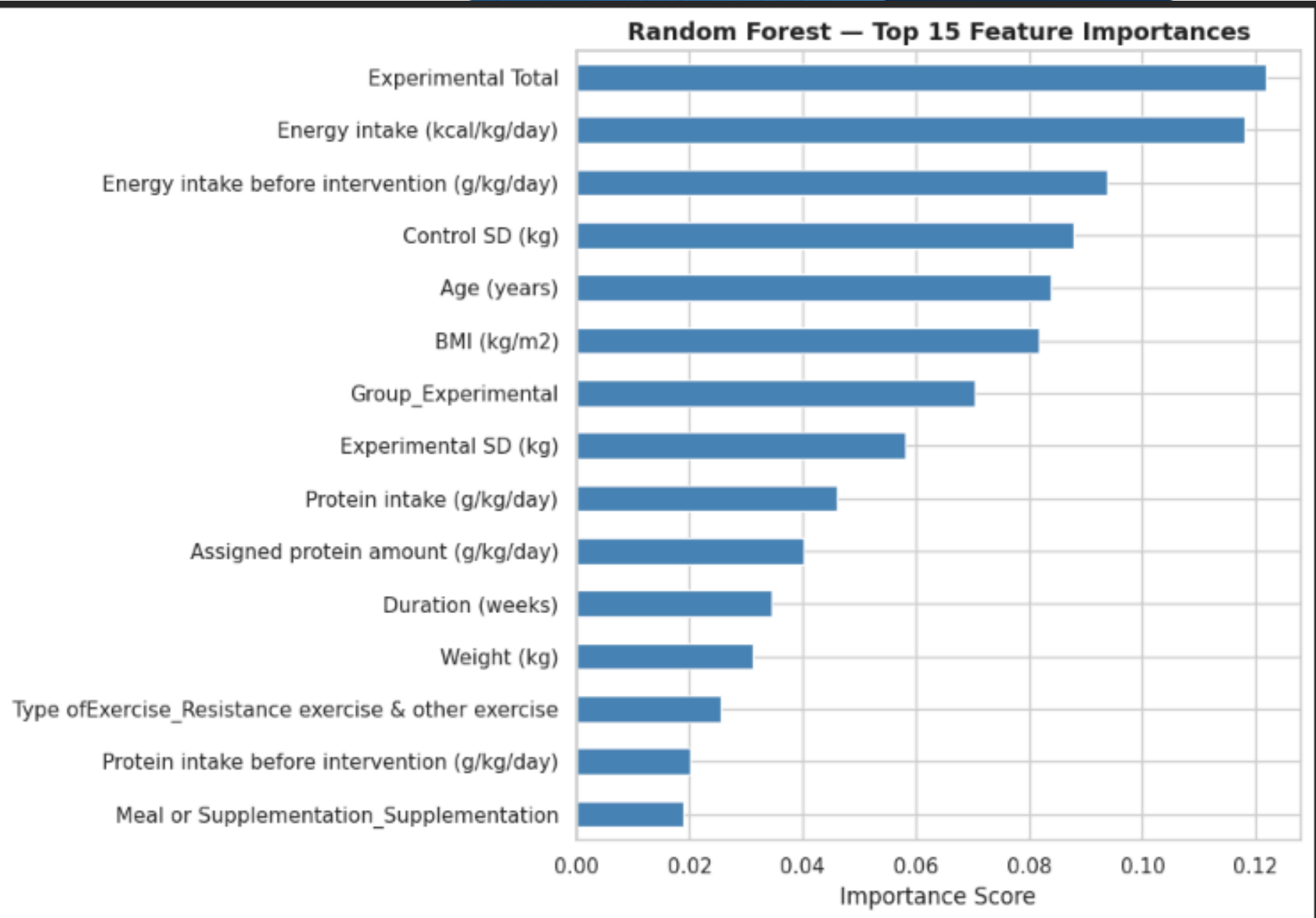
Top 15 most important features:	
Experimental Total	0.1219
Energy intake (kcal/kg/day)	0.1182
Energy intake before intervention (g/kg/day)	0.0938
Control SD (kg)	0.0879
Age (years)	0.0840
BMI (kg/m2)	0.0818
Group_Experimental	0.0703
Experimental SD (kg)	0.0581
Protein intake (g/kg/day)	0.0461
Assigned protein amount (g/kg/day)	0.0402
Duration (weeks)	0.0346
Weight (kg)	0.0313
Type ofExercise_Resistance exercise & other exercise	0.0255
Protein intake before intervention (g/kg/day)	0.0201
Meal or Supplementation_Supplementation	0.0190

Model Interpretation

The feature importance analysis reveals which variables most strongly predict muscle hypertrophy.

Random Forest identifies Duration, Protein Intake, and Training Frequency as top predictors.

When cross-checked by Lasso regression, it confirms these findings, with non-zero coefficients highlighting Energy Intake and BMI as additional contributors.



Recommendations

Based on our analysis,

- Optimal muscle hypertrophy is achieved with protein intake of 1.6-2.2 g/kg/day.
- Combined with resistance training 3-4 times weekly.
- Higher energy intake and longer training duration (8+ weeks), significantly correlate with increased lean body mass gains.



Key Takeaway

Protein + Training = Results



Thank You

Key Takeaways:

Protein intake (g/kg/day) and training frequency are the strongest predictors of muscle hypertrophy.

The Random Forest and Gradient Boosting models achieved the best predictive performance. Future research should explore longitudinal data and individual genetic factors while driving further conclusions.



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Questions & Discussion



GitHub Repo